

1.	Which of the following apparatus/instruments is most suitable for obtaining both cooking oil and water from a mixture of the two?	5,	In descending Group VI of the periodic table, which TWO of the following trends occur?
	(A) Filter funnel (B) Separating funnel (C) Fractionating column (D) Volumetric flask		I. The density of the elements increases II. The relative atomic mass of the elements decreases III. The melting point of the elements increases IV. The reactivity of the elements increases
2.	Which TWO of the following statements are true about the arrangement of electrons, protons and neutrons in an atom?		
	1. Protons' and neutrons are found in the-nucleus.		(A) I and III (B) I and IV (C) II and III (D) II and IV
	II. Electrons can be found anywhere outside the nucleus.	6.	Dilute sulphuric acid is classified as a strong acid because it
	III. The number of protons always equals the number of neutrons.		
	IV. The number of protons always equals the number of electrons.		(A) produces 2 moles of hydrogen ions per mole of acid (B) requires 2 moles of sodium hydroxide for neutralisation (C) is almost completely ionised and is a good conductor of electricity (D) ionises to give both hydrogen ions and sulphate ions.
	(A) I and III (B) I and IV (C) II and III (D) III and IV		

3. Which of the following compounds may be conveniently prepared by precipitation?

- (A) Barium sulphate
(B) Sodium sulphate
(C) Copper (II) sulphate
(D) Magnesium nitrate

4. Which of the atoms represented below by their electronic configuration will most readily form a positive ion?

- (A) 2,8,1
(B) 2,8,2
(C) 2,8,7
(D) **2,8,8**

7. **When a saturated solution of copper (II) sulphate is cooled, crystals of copper (II) sulphate-5-water begin to form because solubility of copper(II) sulphate-5-water**

- (A) increases with decreasing temperature
(B) decreases with increasing temperature
(C) decreases with decreasing temperature
(D) increases with increasing temperature

8. The arrangement of elements in the periodic table is based on

- (A) relative atomic mass
- (B) atomic number
- (C) mass number
- (D) relative molecular mass

9. When iron (III) sulphate reacts with aqueous potassium iodide, a brown colouration of iodine is produced. Which of the following deductions is correct?

- /(A) Iron (III) sulphate is a reducing agent.
- .. (B) The iodide ion, I⁻, has been oxidised to iodine.
- l(C) The iron(III) sulphate has lost electrons.
- (D) The iodide ion, I⁻, has gained electrons.

10. Which of the following statements about sodium metal are CORRECT?

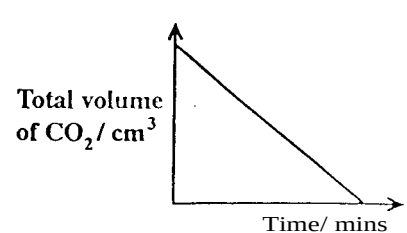
- I. It contains Na ions and mobile electrons.
- II. It contains Na ions and mobile electrons.
- III. It contains cations which repel each other.
- IV. It contains anions which repel each other.

- (A) I and III only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

11. When aqueous silver nitrate is added to an aqueous solution of magnesium chloride, a white precipitate forms. The ionic equation for the formation of this precipitate is

- (A) $\text{Mg}^{2+}(\text{aq}) + \text{NO}_3^{-}(\text{aq}) = \text{MgNO}_3(\text{s})$
- (B) $\text{Ag}^{+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq}) = \text{AgCl}_2(\text{s})$
- (C) $\text{Mg}^{2+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) = \text{MgI}_2(\text{s})$
- (D) $\text{Ag}^{+}(\text{aq}) + \text{Cl}^{-}(\text{aq}) = \text{AgCl}(\text{s})$

12.	In which TWO of the following equations are the UNDERLINED reagents acting as oxidising agents?	
I.	$2\text{H}_2(\text{g}) + \text{SO}_2(\text{aq}) = 2\text{H}_2\text{O}(\text{l}) + \text{S}(\text{s})$	
II.	$\text{I}_2(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) + 2\text{KI}(\text{aq}) = \text{K}_2\text{SO}_4(\text{aq}) + 2\text{HI}(\text{aq}) + \text{I}_2(\text{s})$	
III.	$2\text{FeCl}_2(\text{aq}) + \text{Cl}_2(\text{g}) = 2\text{FeCl}_3(\text{aq})$	
IV.	$\text{C}_6\text{O}(\text{s}) + \text{H}_2(\text{g}) = \text{Cu}(\text{s}) + \text{H}_2\text{O}(\text{l})$	
(A)	I and III	
(B)	I and IV	
(C)	II and III	
(D)	II and IV	

13. When crystals of potassium nitrate are dissolved in water, the temperature of the solution decreases because
- (A) little energy is required to break down the crystal, structure of the potassium nitrate
- (B) heat is always absorbed when a substance dissolves ,
- (C) the energy content of dissolved potassium nitrate is higher than that of solid potassium nitrate
- (D) potassium nitrate is colder than water
14. The high melting point of ionic compounds may be due to the
- (A) arrangement of ions
- (B) large numbers of ions
- (C) attraction among ions
- (D) movement of ions
15. Which of the following **statements about chemical reactions** is NOT correct?
- (A) Energy is given out when bonds break and taken in when bonds form.
- (B) Chemical reactions involve the making and breaking of bonds.
- (C) Endothermic reactions take energy from the surroundings.
- (D) Exothermic reactions give energy to the surroundings.
16. A substance that conducts an electric current but remains chemically unchanged is
- (A) aqueous **copper** (II) sulphate
- (B) copper
- (C) sulphur
- (D) solid sodium chloride
17. Which of the following can exactly neutralise 20 cm³ of 2.0 mol dm⁻³ sodium hydroxide?
- I. 10 cm³ of 4.0 mol dm⁻³ hydrochloric acid
- II. 10 cm³ of 2.0 mol dm⁻³ sulphuric acid
- III. 20 cm³ of 4.0 mol dm⁻³ nitric acid
- (A) I only
- (B) I and II only**
- (C) I and III only
- (D) II and III only**
18. Some **calcium** carbonate was reacted with excess dilute hydrochloric acid. The volume of carbon dioxide evolved was recorded and plotted against time. Which of the following graphs represents this reaction?
- (A)
- Total volume of CO₂/cm³
- Time/ mins
- (B)
- Total volume of CO₂/cm³
- Time/ mins
- (C)
- Total volume of CO₂/cm³
- 
- Time/ mins
- (D)
- Total volume of CO₂/cm³
- Time/ mins

19. In which of the following processes is fractional distillation NOT used?

(A)	Refining of crude petroleum	I.
(B)	Separation of methanol from a methanol-water mixture	II.
(C)	Separation of chlorophyll in a leaf extract	III.
(D)	Separation of liquid air into nitrogen and oxygen	IV.
20.	Graphite can be used as a lubricant because of	(A) (B) (C) (D)

22. The mass of an element produced in electrolysis is proportional to the

quantity of electricity which has been passed
volume of the electrolyte
mass of the electrolyte
mass of the electrode

- I only
II and IV only
I and III only
III and IV only

(A)	strong attraction between the hexagonal layers of carbon atoms	23.	Which of the following statements BEST characterises a catalyst?
(B)	weak attraction between the hexagonal layers of carbon atoms	(A)	It increases the activation energy of the reaction.
(C)	the loose electrons which can move throughout the lattice	(B)	It alters the quantity of the products formed.
(D)	strong attraction within the hexagonal layers of carbon atoms	(C)	It is always unchanged physically at the end of a reaction.
21.	Which of the following sulphates are insoluble?	(D)	It is always unchanged chemically at the end of a reaction.
I.	Barium sulphate	24.	Which of the following operations results in sublimation?
II.	Magnesium sulphate	(A)	Heating of solid sodium chloride
III.	Lead sulphate	(B)	Cooling of oxygen gas
IV.	Aluminium sulphate	(C)	Heating of ethanol
(A)	I and III only	(D)	Heating of iodine crystals
(B)	I and H only		
(C)	II and IV only		
(D)	III and IV only		

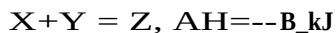
25. How many neutrons and electrons does the

particle	$^{24}_{12}\text{X}^{2+}$	have?	
	Neutrons		Electrons
(A)	12		10
(B)	12		12
(C)	24		10
(D)	24		12

26. Which of the following sources of energy is the MOST commonly used in the world today?

- (A) Nuclear
- (B) Fossil fuels
- (C) Biogas
- (D) Gasohol

27. Study the following thermochemical equation.



Which of the following methods can be used to compute the value of $-B$ kJ?

- I. Energy of Z minus the energy of X and Y
- II. Energy of X and Y minus the energy of Z
- III. The sum of the energies of X, Y and Z

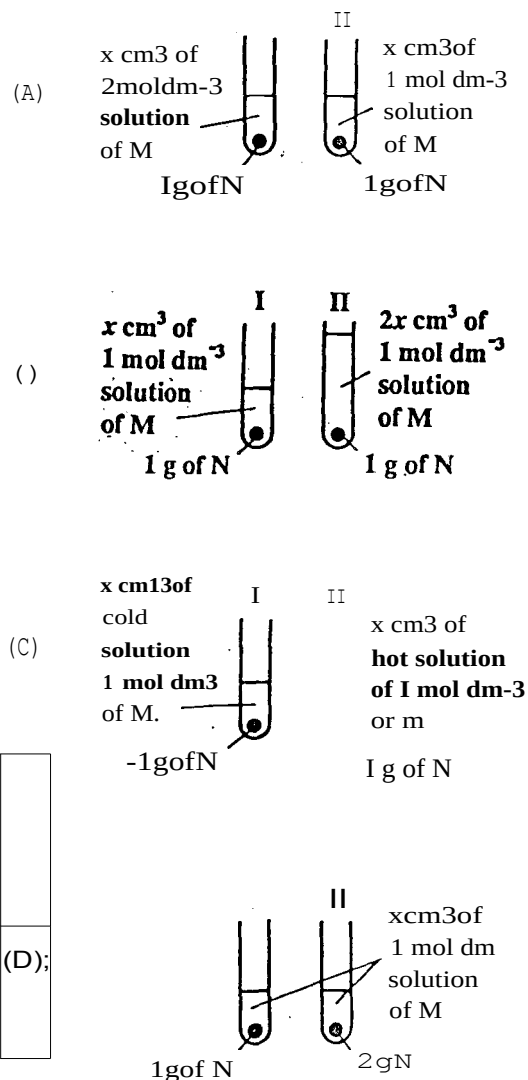
- (A) I only
- (B) I and II only
- (C) III only
- (D) II and III only

28.	Which of the following processes provides evidence in support of the particulate nature of matter?	
	(A) Diffusion (B) Precipitation (C) Condensation (D) Filtration	(D);

29. Sulphuric acid forms the sodium salts NaHSO_4 and Na_2SO_4 . Its basicity is therefore

- (A) 1
- (B) **2**
- (C) 3
- (D) 4

30. A sample of substance N was placed in solution M. Which of the following diagrams represents pairs of experiments that would be most suitable to determine how the concentration of M affects the rate of the reaction?



31. Atoms of **isotopes**, X and Y, have
- I. the same number of electrons..
 - II. the same number of protons.
 - III. different numbers of neutrons.
- (A) III only
(B) II and III only
 (C) I and II only
 (D) I, II and III

35.	All of the following gases are colourless EXCEPT	
	(A)	O ₂
	(B)	
	(C)	NO ₂
	(D)	NH ₃

32.	Which of the following could produce carbon dioxide and carbon monoxide as pollutants?		36.	Which of the following elements reacts most vigorously with I ⁻ (aq) ions to give hydrogen gas?	
	I.	Geothermal energy		(A)	Zinc
	II.	Solar energy		(B)	Lead
	III.	Fossil fuels		(C)	Iron
	IV.	Biogas		(D)	Copper
	(A)	I only			
	(B)	II only			
	(C)	II and III only	37.	Which of the following is a weak electrolyte?	
	(D)	III and IV only			

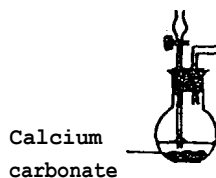
33. At 1 atmospheric pressure, the particles move fastest in a
- (A) liquid at 25 °C
 (B) gas at 25 °C
 (C) liquid at 100 °C
 (D) gas at 100 °C

- (A) Molten lead
 (B) Aqueous ethanoic acid
 (C) Dilute hydrochloric acid
 (D) Molten lead bromide

34. Which of the following pairs of acids is monobasic?
- (A) HCl and CH₃CO₂H
 (B) HCl and H₂SO₄
 (C) HNO₃ and H₂SO₄
 (D) CH₃CO₂H and H₂SO₄

Item 38 refers to the following diagrams.

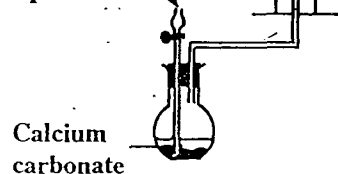
Dilute sulphuric acid \



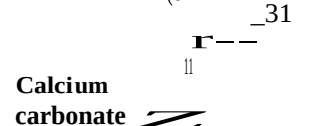
11 Dilute hydrochloric acid



III Dilute sulphuric acid



IV Dilute hydrochloric acid



38.	In which of the experiments would the amount of carbon dioxide collected be GREATEST?			
(A)	I			
(B)	II			
(C)	III			
(D)	IV			3.

39.	Which of the following gives an alkaline reaction with moist litmus paper?			(A)	Electrolysis of molten chloride
				(13)	Electrolysis of aqueous chloride
	(A)	Ammonia		(C)	Electrolysis of molten oxide
	(B)	Nitrogen (IV) oxide		(D)	Reduction of oxide with carbon
	(C)	Hydrogen chloride			
	(D)	Water			

40. Which of the following aqueous solutions will produce blue precipitate with aqueous sodium hydroxide?

- (A) Calcium nitrate
- (B) Iron (II) nitrate
- (C) Copper (II) nitrate
- (D) Aluminium nitrate

41. Which of the following statements is true?

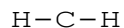
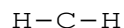
- (A) Ammonia is an acidic gas.
- (B) Ammonia is soluble in water and functions as a base.
- (C) Nitric acid is used as a fertilizer and a reducing agent.
- (D) Nitric acid contains two replaceable hydrogen atoms.

42. Aluminium is more reactive than iron yet, after a while, a piece of aluminium left exposed to the atmosphere becomes corrosion resistant whilst a similar piece of iron continues to corrode. This is because the

- (A) aluminium has reacted completely
- (B) iron oxide formed is more reactive than the aluminium oxide formed
- (C) oxide film on aluminium protects the metal from further corrosion but the oxide film on iron does not
- (D) iron oxide reacts with the iron below but aluminium oxide does not react with the aluminium below

Which of the following methods is used for the extraction of metals at the top of the electrochemical series?

			c 4
		9	
44.	Which of the following metals will NOT react with water under any conditions?	48.	A compound has the following structural formula.
(A)	Magnesium		$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ & & & \\ \text{H} & \text{H} & & \text{H} \end{array}$
(B)	Aluminium		
(C)	Iron		
(D)	Copper		



45. Which of the following observations would you expect to make when excess sodium hydroxide **solution** is added to a solution containing **zinc ions**?

Which TWO of the following statements are correct? The **compound**

(A)	An alkaline gas produced	I..	is an alcohol.
(B)	White precipitate soluble in excess	II.	is a branched alkane.
(C)	White precipitate insoluble in excess	III.	can react with sodium.
(D)	A green precipitate insoluble in excess	IV.	Is an unreactive substance.
Items 46-47 refer to the following organic compounds.		(A)	I and III
		(B)	I and IV
		(C)	II and III
		(D)	II and IV
(A)	Ethanol		
(B)	Ethene		
(C)	Ethanoic acid		
(D)	Ethyl ethanoate	49.	Which of the following would you expect to form an addition polymer?

In answering **items**. 46-47 a particular choice from the above maybe made more than once, once, or not **at all**.

Which of the **organic compounds**

46. undergoes addition reactions?

47. is immiscible with water and is sweet-smelling?

(A)	$\text{CH}_2=\text{CH}$ CN
(B)	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{NH}_2 \end{array}$
(C)	$\begin{array}{c} \text{O} \\ \\ \text{C}-\text{OH} \end{array}$ C ₁₇ H ₃₅
(D)	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$

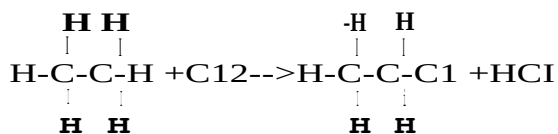
50. **Ethanol** is NOT used as a

- (A) beverage
- (B) fuel
- (C) lubricant
- (D) solvent

51. In which of the following is the amide linkage present?

- (A) Polythene
- (B) Nylon
- (C) Starch
- (D) Terylene

52.



The equation shown above represents a reaction which is classified as

- (A) addition
- (B) neutralisation
- (C) polymerisation
- (D) substitution

53. **Yeast** may be used in the fermentation of starch to ethanol because

- (A) it is acidic
- (B) it contains enzymes which act on starch
- (C) yeast itself produces the ethanol
- (D) it fixes the oxygen needed in the reaction from the atmosphere

54. The process by which large molecules of hydrocarbon are broken up into smaller molecules is called

- (A) saponification
- (B) cracking
- (C) polymerization
- (D) condensation

55. Which of the following is/are condensation polymers?

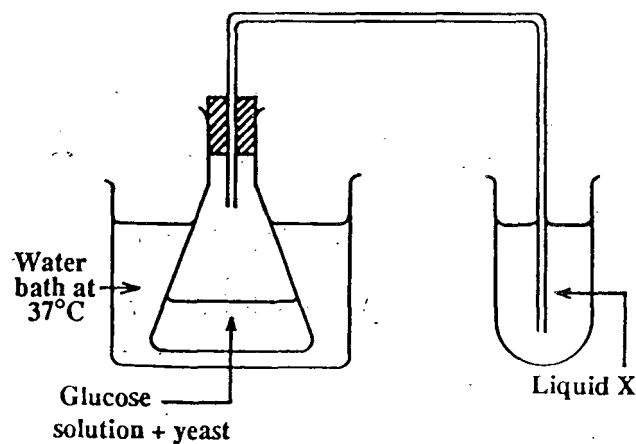
- I. Terylene
- II. Nylon
- III. Starch
- IV. Polythene

- (A) I only
- (B) I and IV only
- (C) I, II and III only
- (D) II, III and IV only

56. Which of the following substances gives a blue colouration with iodine?

- (A) Fats
- (B) Starches
- (C) Proteins
- (D) Polyamides

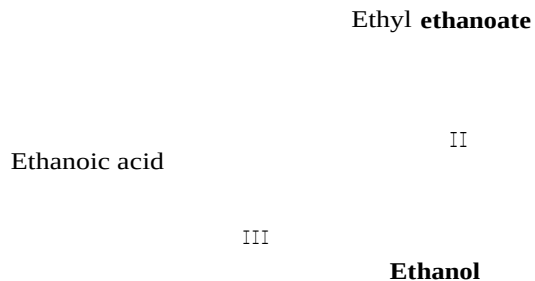
Item 57 refers to the following diagram of an experiment.



57. The gas produced can be BEST identified if Liquid X is

- (A) lime water
- (B) litmus solution
- (C) bromine water
- (D) sodium hydroxide solution

Items 58 - 59 refer to the following diagram which represents three conversions in organic chemistry



58. Which of the following equations BEST represents Conversion I?

- (A) $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$
- (B) $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_5\text{COOCH}_3 + \text{H}_2\text{O}$
- (C) $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$
- (D) $\text{C}_2\text{H}_5\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_5\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$

59. For Conversion II, ethyl ethanoate is heated under reflux with another compound. The name of the compound is

- (A) sodium chloride
- (B) ethanol
- (C) fatty acids
- (D) sodium hydroxide

60. The compound propene, $\text{CH}_3\text{CH}=\text{CH}_2$, reacts with bromine to produce

- (A) $\text{CH}_3\text{CHBrCH}_2\text{Br} + \text{HBr}$
- (B) $\text{CH}_3\text{CHBrCH}_2\text{Br} + \text{HBr}$
- (C) $\text{CH}_3\text{CHBrCH}_2\text{Br}$
- (D) $\text{CH}_3\text{CHBrCH}_2\text{Br}$

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.